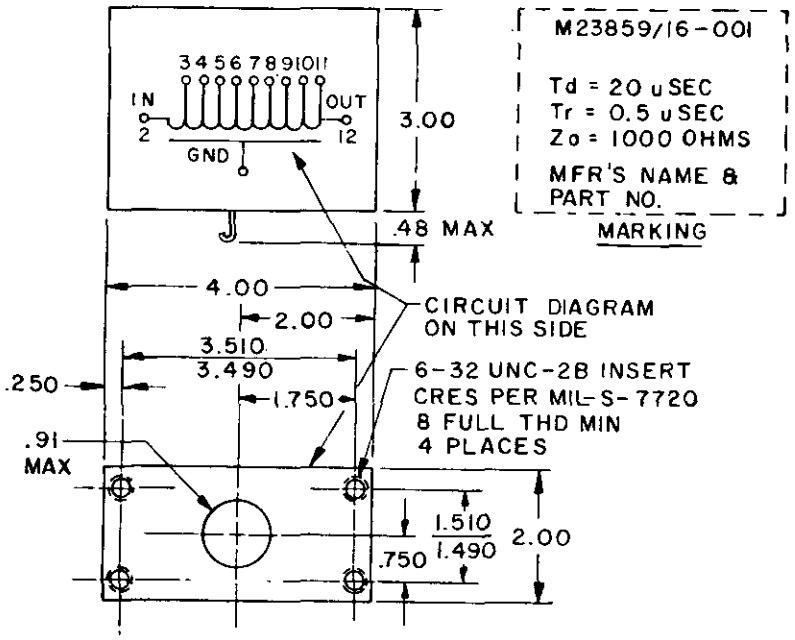
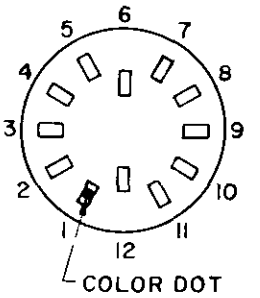
MILITARY SPECIFICATION SHEET

DELAY LINE, PULSE, ELECTROMAGNETIC, LUMPED CONSTANT, HERMETICALLY SEALED (2.0 - 20.0 μsec DELAY)

The complete requirements for procuring the delay lines described herein shall consist of this document and the latest issue of Specification MIL-D-23859.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.





TERMINAL MARKING

INCHES	MM	INCHES	MM
.250	6.35	1.750	44.45
.48	12.19	2.00	50.80
.750	19.05	3.00	76.20
.91	23.11	3.490	88.65
1.490	37.85	3.510	89.15
1.510	38.35	4.00	101.60

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.03 (.76 mm) and ±.015 (.38 mm).

FIGURE 1. Case and mounting dimensions.

DATA.

Part number	Nominal terminating impedance (ohms)	Delay (μsec)		Attenuation (db)	Rise time at max delay (μsec)
		Tap No	Delay ±0.2		
M23859/16-001	1,000	2-3	2.0	25	.5
		2-4	4.0		
		2-5	6.0		
		2-6	8.0		
		2-7	10.0		
		2-8	12.0		
		2-9	14.0		
		2-10	16.0		
		2-11	18.0		
		2-12	20.0		

FSC 5840

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ENGINEERING DATA:

Case material: Ferrous type, hot tinned or fuzed electrotinned.

Finish: Grey enamel in accordance with MIL-E-15090, type I, class 2.

Weight: Not applicable.

Terminals: Twelve hooked type solder lugs capable of accepting two No. 22 AWG buss leads minimum.

Input pulse:
Peak pulse voltage: 150 volts peak maximum.
Pulse width: Shall be greater than three times the delay line pulse rise time and less than twice the delay.
Pulse rise time: 15 percent to 20 percent of delay line rise time.

Nominal terminating impedance: See table.

Style: Not applicable.

Figure of merit: Not applicable.

REQUIREMENTS:

Design and construction: See figure 1.

Enclosure: See figure 1.

Delay time: See table.

Rise time: See table.

Attenuation: See table.

Distortion: ±8 percent at maximum delay.

Dielectric withstanding voltage:
Method 301 of MIL-STD-202.

Delay time variation with temperature:
±0.5 percent from -55°C to +75°C.

Insulation resistance: Method 302, test condition A, MIL-STD-202. 1,000 megohms minimum.

Thermal-age stability: 3 percent maximum delay change permissible. All other electrical characteristics shall still meet the requirements of this specification.

Sealing: As specified in MIL-D-23859.

Terminal strength: Method 211 of MIL-STD-202.
Test condition A - Applied force 4-1/2 pounds.
Test condition C - Applied force 5 pounds.

Salt atmosphere: Method 101, test condition B, MIL-STD-202.
Salt solution - 5 percent.
Measurements - Not applicable.

Pressure: Method 105, test condition C, MIL-STD-202. Perform dielectric withstanding voltage test during reduced pressure and at 250 vdc.
Method of mounting - Normal.

Vibration: Method 204, test condition B, MIL-STD-202.
Measurements - Not applicable.

Shock: Method 205, test condition A, MIL-STD-202.

Thermal shock: Method 107, test condition A, MIL-STD-202, except that upper temperature limit is 95°C.
Measurements - Dielectric withstanding voltage and insulation resistance.

Immersion: Method 104, test condition A, MIL-STD-202.
Measurements - Insulation resistance.

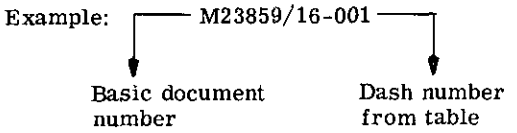
Moisture resistance: Method 106 of MIL-STD-202.
Loading voltage - Not applicable.
Polarizing voltage - 100 volts dc.
Measurements - Insulation resistance and dielectric withstanding voltage.

Flammability: Not applicable.

Life: Method 108, test condition D, MIL-STD-202.
Test temperature - 75° ±2°C.
Test voltage - 150 vdc applied.
Measurements - Dielectric withstanding voltage, insulation resistance, delay time variation with temperature, and thermal age stability.

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Part number: Consists of the basic number of this specification sheet and a dash number taken from table.



Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU, EL
Navy - EC, AS, OS
Air Force - 80, 17, 85
DSA - ES

User activities:
Army - MI
Navy - MC, CG
Air Force -

Preparing activity:
Army - EL

(Project 5840-0044)

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